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# Moayed Haji Ali

Machine Learning Engineer

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## Education

**Rice University** — **Houston, TX**
*PhD in Computer Science* — *Aug 2023 - June 2027 (expected)*

**Koç University** — **Istanbul, Turkey**
*Bachelor in Computer Engineering, CGPA:* ***3.96/4.0****, Major GPA:* ***3.98/4.0*** — *Jan 2019 - June 2023*

- Ranked **3rd** in the Computer Science Department
- Full scholarship merit and need based from the Al Ghurair Foundation for Education (AGFE)

**AlSaade High School,** CGPA: 94.1 — **Damascus, Syria;** 2017-2018

## Publications

- [1] **M. H. Ali**, A. Bond, T. Birdal, D. Ceylan, L. Karacan, E. Erdem, and A. Erdem "VidStyleODE: Disentangled Video Editing via StyleGAN and NeuralODEs". Accepted at **ICCV 2023**
- [2] Y. H. Nazarabadi, **M. H. Ali**, and N. Nayal "Opera: Scalable Simulator for Distributed Systems". [Demo Paper]. Accepted at **IEEE INFOCOMP 2021**.

## Relevant coursework

- Autonomous Driving
- Advances in Deep learning
- Deep Unsupervised Learning
- Deep Learning and Computer Vision
- Introduction to Machine Learning
- Natural Language Processing
- Database Management
- Algorithms and Complexity
- Software Engineering Design

## Awards

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|---|---|
| Seven times recipient of the Vehbi Koç Scholar Award given for outstanding classwork performance | 2019 - 2022 |
| Syrian Youth Empowerment grant; **1000 USD** | 2022 |
| Best project of the year award in the "Software Engineering Design" course | 2020 |
| Flight and accommodation grant for Best short talk video at AGFE Summit 2019 | 2019 |
| Honorable mention in Asia-Pacific Olympiad of Informatics | 2016 |

## Experience

**Remote Machine Learning Engineer** — **Feb 2021 - Jan 2022, April 2023 - Aug 2023**
*CareX* — *California, USA*

- Worked on the development of a blood pressure estimation model from face videos using Pytorch
- Implemented a Pytorch and Tensorflow framework for building, training, and visualizing deep learning models
- Contributed to HRV estimation model from fingertip videos technology, which was deployed in CareX's SDK

**Computer Vision Research Intern** — **Aug 2021 - July 2023**
*Koc University and Is Bank Artificial Intelligence Center (KUIS-AI Center)* — *Istanbul, Turkey*

- Leading an ongoing project on unsupervised disentangled image representation using multimodal neural networks
- Lead a research project on video modeling and manipulation using StyleGAN2 and Latent ODEs.
- Implemented using Pytorch a baseline for text-guided image manipulation based on CLIP and TokenGAN
- Improved upon a recent variational autoencoder-based architecture (VQGAN) using language supervision

**Programming Tutor** — **July 2021 - June 2023**
*Hub21* — *Istanbul, Turkey*

- Delivered +200 hours of private tutoring in programming using Python, JAVA, and C++

**Remote AI Researcher** — **Aug 2022 - Sep 2022**
*King Abdullah University of Science and Technology* — *Thuwal, Saudi Arabia*

- Conducted and summarized an extensive literature review on text-guided 3D generation

**Undergraduate Research Intern** **June 2020 - June 2021**
*Intelligence User Interface Lab (IUI)* *Istanbul, Turkey*

- Designed and implemented using Tensorflow an intelligent feedback system for sketch answers to school questions.
- Technologies included object-level segmentation, image registration, non-smooth function optimizations, prototype-based object recognition, and planer graph morphing.

**Undergraduate Research Intern** **June 2019 - Oct 2019**
*Distributed Systems and Reliable Networks Research Laboratory (DISNET)* *Istanbul, Turkey*

- Implemented a distributed computer network simulator using Java that was published as a demo paper at IEEE INFOCOM 2021 conference

## Teaching Experience

**Deep Unsupervised Learning (Combined Graduate/Undergraduate Course)** **Feb 2022 - June 2022**

- Prepared an assignment on Diffusion Models and contributed to grading the course assignments
- Assisted students with their course projects and held weekly office hours

**Advanced programming in java (Undergraduate Course)** **Sep 2020 - Jan 2021**

- Conducted weekly lab sessions and graded students' assignments

**Koc University Office of Learning and Teaching (KOLT)** **Jan 2020 - May 2020**

- Held office hours, review sessions, and proctored exams for the "Introduction to Programming" course

## Technical Projects

**Coolingo** **Spring 2022**
*An Android bot for teaching children English interactively*

- Developed a video chat-bot that combines several SOTA models effectively: BlenderBot 2.0, Google T5, MakeItTalk, FastSpeech2.0, and GPT3
- Deployed AI models on AWS lambda, EC2 instance, and a local server.

**Video Implicit Representation** **Spring 2021**
*Representing videos' temporal dimension as an implicit function parameterized by a transformer architecture*

- Replicated VideoGPT implementation using Pytorch and implemented a novel network extension.

## Skills

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|---|---|
| **Programming Languages** | Python, Java, C++, Julia, MYSQL, HTML, CSS, and React |
| **Technologies** | Git, LaTeX, Android Development, Web development, restful API, AWS, and Docker |

## Competitions

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|---|---|
| Huawei Koç university mobile development competition; ranked **1st** | 2020 |
| Inzva online programming competition; ranked **5th** | 2020 |
| Startup Roadshow 2019 Istanbul; ranked **1st** | 2019 |
| LEAP KWorks Accelerator Startup Competition; ranked **2nd** | 2019 |
| IEEEXtreme Programming Contest; ranked **2nd** in Turkey | 2019 |
| Turkish Programming Contest; ranked **2nd** | 2019 |
| Damascus ACM College Programming Contest; ranked **1st** | 2015 & 2016 |

## Extra-curricular activities

**Co-founder of Yders.com** **2019 - 2021**

- Co-founded Yders, a platform that empowers university students and connects them with K-12 students to assist them with their studies
- Represented Yders in Startup Roadshow finals in Amman, Jordan

**Team member of the Syrian Olympiad Team Of Informatics (SOI)** **2015 - 2017**

- Attended three two-month long programming bootcamps
- Represented the Syrian team in the Asia-Pacific Informatics Olympiad